DATA MINING PROJECT REPORT

TOPIC: “IMPACT OF IMBALANCED DATA ON MODEL ACCURACY”

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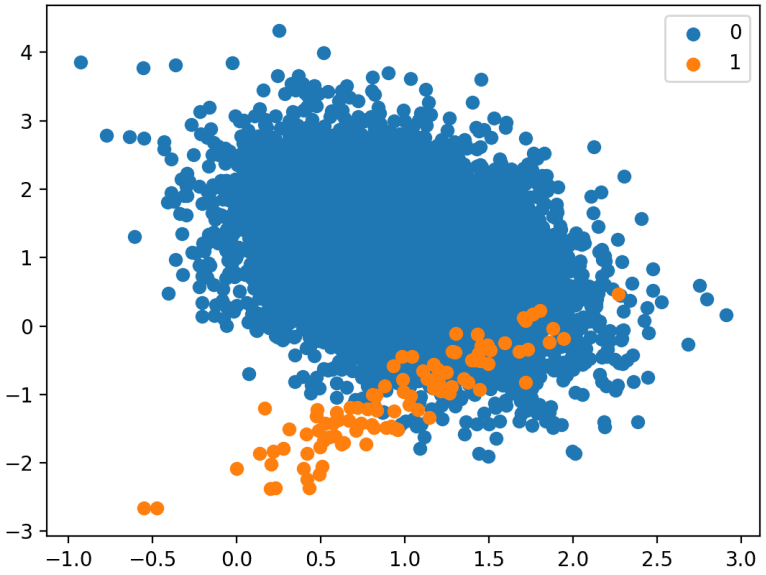
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1. Introduction to Imbalanced Data

Imbalanced data refers to a dataset where the target classes are not equally represented. In most cases, one class (the **majority class**) contains a significantly larger number of instances than the other class (the **minority class**). This situation is very common in real-world applications, because many important events are naturally rare, expensive to collect, or difficult to label.

When training machine learning models on imbalanced datasets, models often become biased toward predicting the majority class, since it dominates the learning process. As a result, the model may appear "good" using simple accuracy, while actually failing at the task we truly care about: correctly identifying minority cases. This is especially problematic when minority cases are critical, such as fraud detection, rare disease detection, or employee attrition prediction.



1.1 What is Imbalanced Data?

A dataset is considered imbalanced when the distribution of class labels is skewed. For example, if 95% of samples belong to class 0 and only 5% belong to class 1, the model can achieve 95% accuracy by always predicting class 0 - yet it would detect none of the minority events. This leads to biased outcomes and unreliable decision-making in applications where identifying minority cases is important.

1.2 Real-World Examples of Imbalanced Data

Imbalanced datasets occur in many domains:

- **Healthcare & Rare Diseases:** Diseases such as cancer are rare compared to "no cancer" cases. Even though positive cases are few, detecting them is crucial for early intervention and saving lives.
- **Fraud & Anomaly Detection:** Fraudulent transactions are far less frequent than legitimate ones. Missing fraud can lead to major financial loss.
- **Customer Churn & Risk Analysis:** Only a small portion of customers typically leave (churn). Predicting these rare cases helps businesses retain customers and reduce revenue loss.

2. Causes and Origins of Imbalance

Class imbalance usually originates from how data is collected, from the nature of the events being modeled, or from mistakes/bias introduced in data preparation. Understanding these causes is important because it helps choose the right handling techniques and avoids creating misleading models.

2.1 Data Collection Issues

2.1.1 Sampling Bias

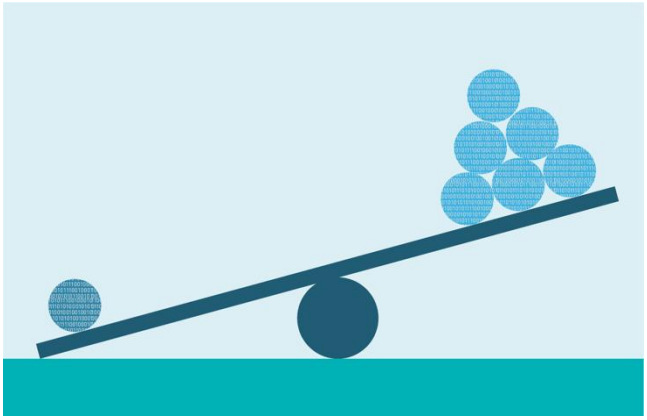
Sampling bias occurs when collected data does not represent the full population fairly. This can happen due to data being collected from limited locations, certain demographic groups, or only specific time periods. When sampling bias exists, some classes become overrepresented while others become underrepresented. The model then learns patterns that work well for majority data but fail on minority cases. Ensuring representative sampling reduces bias and improves reliability.

2.1.2 Data Scarcity

Data scarcity means there are very few minority-class samples available. This happens because the event is rare, expensive to measure, limited by privacy (e.g., medical data), or difficult to observe. With too few minority examples, a model cannot learn stable patterns and tends to misclassify minority cases. Common solutions include data augmentation, synthetic data generation, and oversampling to increase minority representation.

2.2 Natural Rarity of Events

Many minority events occur infrequently by nature: equipment failures, natural disasters, rare diseases, or uncommon genetic mutations. Even with excellent data collection, the minority class remains small because real-world occurrence is low. This naturally creates imbalance and makes it difficult for models to learn minority patterns, often reducing recall and F1-score for the minority class.



2.3 Labeling and Biases

2.3.1 Labeling Errors

Labeling errors occur when data points are labeled incorrectly or inconsistently. This can happen from human mistakes, ambiguous definitions, or differences in labeling guidelines. In imbalanced problems, labeling errors are even more damaging because minority samples are already scarce—so mislabeling a small number of minority cases can heavily distort learning. Proper label validation and cleaning improves model trustworthiness.

2.3.2 Preprocessing Bias

Preprocessing steps (feature selection, normalization, transformations) can unintentionally amplify imbalance effects. For example, removing features that are important for the minority class can make the model "blind" to minority signals. Also, transformations applied unevenly across groups can introduce systematic bias. Careful evaluation of preprocessing is necessary to keep the model fair and accurate for both classes.

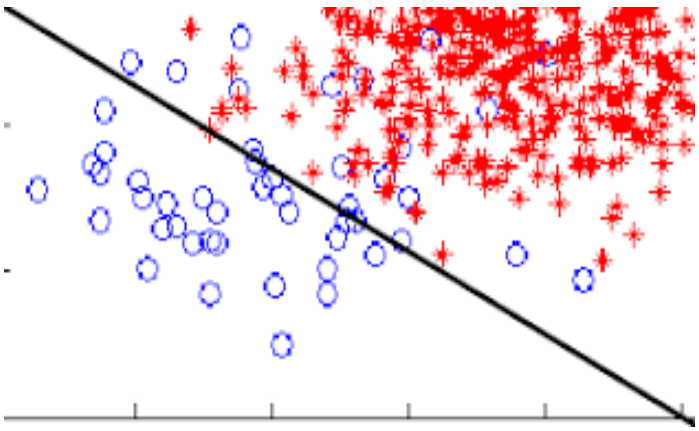
3. Impact on Machine Learning

Imbalanced datasets directly affect how models learn, what patterns they prioritize, and how they generalize to new data. The major issue is that the model becomes strongly influenced by the majority class.

3.1 Bias Toward the Majority Class

3.1.1 Skewed Predictions

Because the model sees many more majority examples, it learns to predict the majority class most of the time. This creates skewed predictions: the model appears strong overall but performs poorly on the minority class. In practice, this leads to low minority recall and low minority F1-score.



3.1.2 Example Scenario

For instance, in credit risk, if approvals dominate the training data, a model may approve most applicants and miss high-risk cases. This shows how imbalance can produce dangerous "optimistic" models that look accurate but fail where it matters.

3.2 Poor Minority Class Learning

3.2.1 Difficulty in Pattern Recognition

Minority patterns are often overshadowed during training, so the model struggles to recognize features that truly differentiate minority cases. This results in misclassification of minority instances and reduces overall effectiveness, particularly in high-stakes tasks like fraud detection or diagnosis.

3.2.2 Lack of Sensitivity

Models become insensitive to minority variations. Even when minority cases differ meaningfully, the model does not learn these distinctions because majority data dominates the loss function and training updates. This produces more false negatives, lowering recall and F1-score.

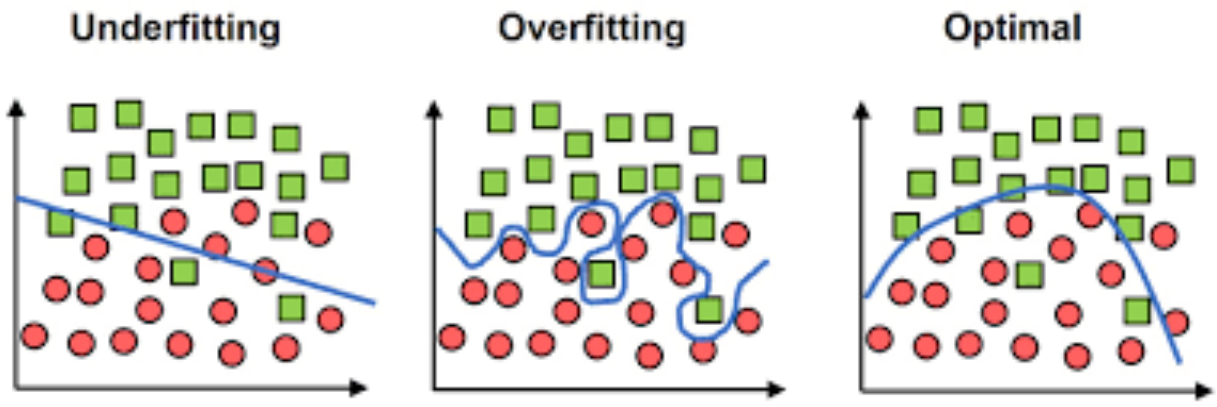
3.3 Overfitting and Generalization

3.3.1 Overfitting to Majority Class

A model may overfit to majority patterns, learning rules that work well on dominant cases in the training set but not in real, diverse scenarios. This can create a false sense of reliability.

3.3.2 Generalization Errors

When minority patterns are not learned correctly, performance drops on new data. The model is unreliable in real-world deployments, especially for rare events, leading to incorrect decisions in fraud, healthcare, or employee attrition prediction.



4. Model Evaluation Challenges

Evaluating models on imbalanced data requires more than accuracy. The main challenge is that standard metrics can hide poor minority performance.

4.1 Limitations of Accuracy

Accuracy can be misleading because a model can achieve high accuracy simply by predicting the majority class most of the time. In an imbalanced dataset, this does not mean the model is useful - especially when the goal is detecting minority cases.

4.2 Precision, Recall, and F1-Score

4.2.1 Precision

Precision measures the correctness of positive predictions: among all predicted positives, how many are actually positive. High precision means fewer false alarms, which is important when false positives are costly.

4.2.2 Recall

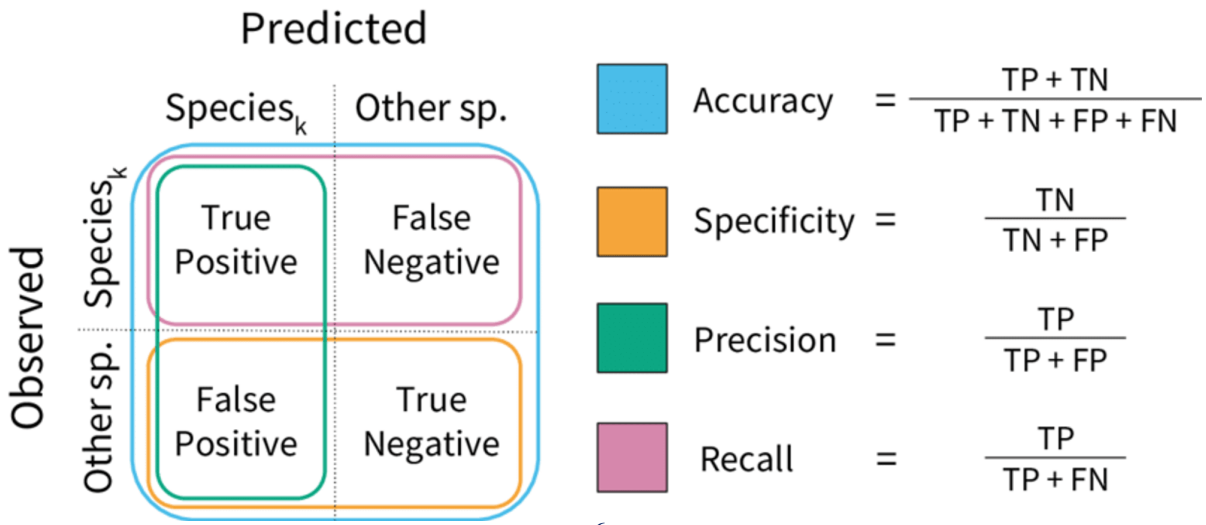
Recall measures how many actual positives were correctly detected. In imbalanced data, recall is critical because missing minority cases often has serious consequences (false negatives).

4.2.3 F1-Score

F1-score is the harmonic mean of precision and recall. It is especially useful for imbalanced datasets because it balances two competing priorities:

- we want to detect minority cases (high recall),
- but we also want predictions to be reliable (high precision).

F1-score provides a single measure that reflects minority-class performance better than accuracy, and it prevents the model from being judged as “good” when it simply predicts the majority class.



5. Consequences of Ignoring Imbalance

Ignoring class imbalance can lead to serious real-world harm. This is because the model systematically misses rare but important events.

5.1 Healthcare Risks

- **Misdiagnosis:** Failing to detect rare diseases delays treatment and increases health risks.
- **Incorrect treatment:** Wrong predictions can lead to treatments that do not match patient needs, worsening outcomes.

5.2 Financial Loss

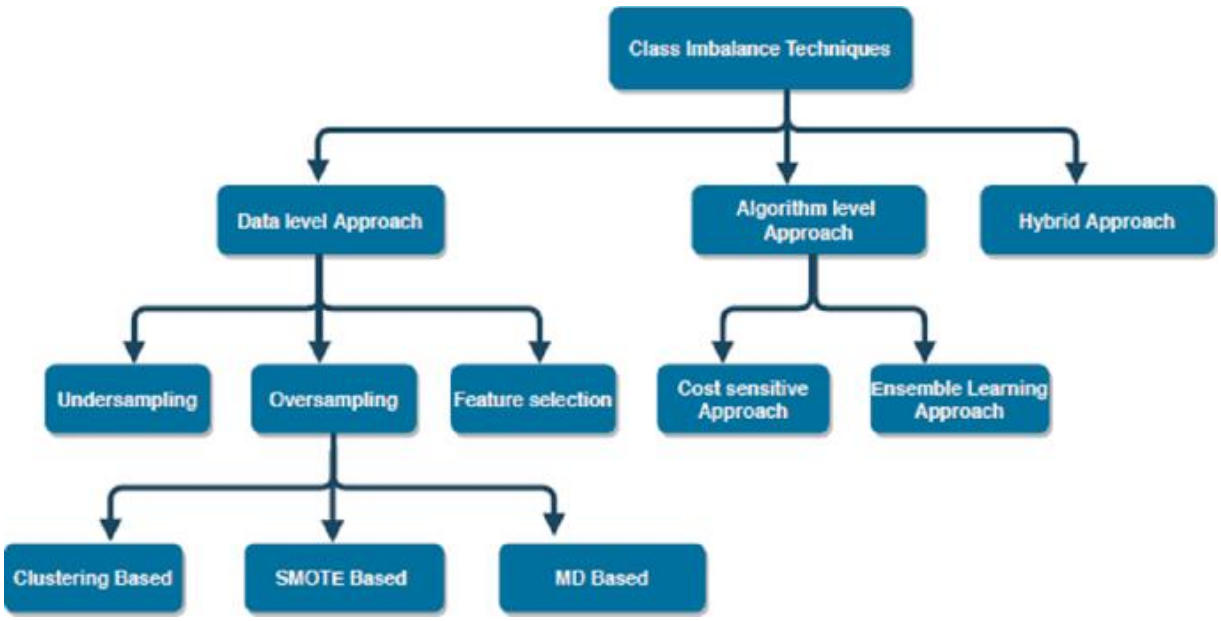
- **Undetected fraud:** Missed fraudulent transactions can create significant financial loss.
- **Regulatory fines:** Poor fraud detection may cause compliance failures, resulting in penalties.

5.3 Safety and Ethical Concerns

- **Safety hazards:** In automated systems, missing rare failures can cause accidents or dangerous incidents.
- **Ethical implications:** Bias in automated decision systems can reinforce unfairness and discrimination.

6. Techniques for Handling Imbalanced Data

To improve minority class detection and build reliable models, imbalance can be addressed using data-level, algorithm-level, hybrid, and advanced approaches.



6.1 Data-Level Approaches

Data-level methods modify the dataset distribution before training.

6.1.1 Resampling Overview

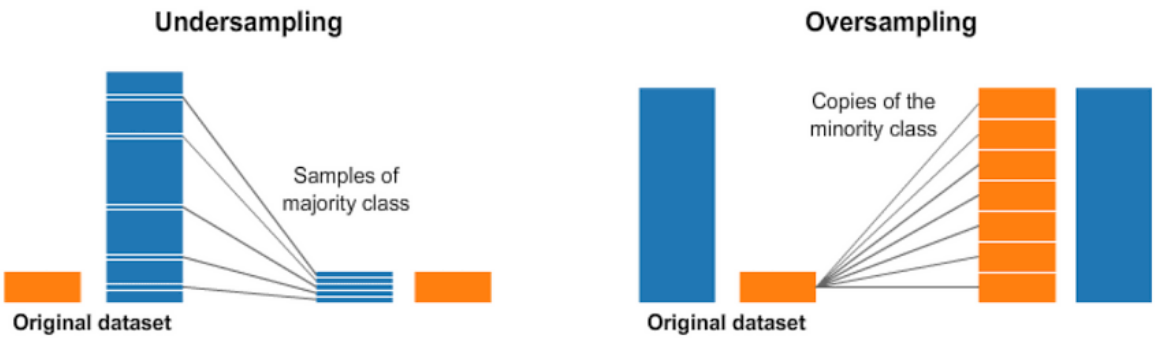
Resampling balances the dataset by either increasing minority samples (oversampling), reducing majority samples (undersampling), or combining both. It helps reduce majority bias and allows models to learn minority patterns, but must be used carefully to avoid overfitting or information loss.

6.1.2 Oversampling Techniques

- **SMOTE:** Generates synthetic minority samples by interpolating between real minority instances, improving generalization compared to simply duplicating samples.
- **ADASYN:** Generates more synthetic samples in difficult regions where minority cases are harder to learn, improving boundary learning.
- **Considerations:** Oversampling can lead to overfitting if synthetic samples become too similar or if oversampling is excessive.

6.1.3 Under-sampling Techniques

- **Random Under-sampling:** Randomly removes majority samples to balance the dataset; simple but may discard useful information.
- **NearMiss:** Keeps majority samples closest to minority samples, aiming to preserve the most informative majority cases.
- **Tomek Links:** Removes pairs of points from different classes that are very close, cleaning boundaries and reducing overlap.



6.2 Algorithm-Level Approaches

Algorithm-level methods keep the dataset unchanged but modify training behavior.

6.2.1 Cost-Sensitive Learning

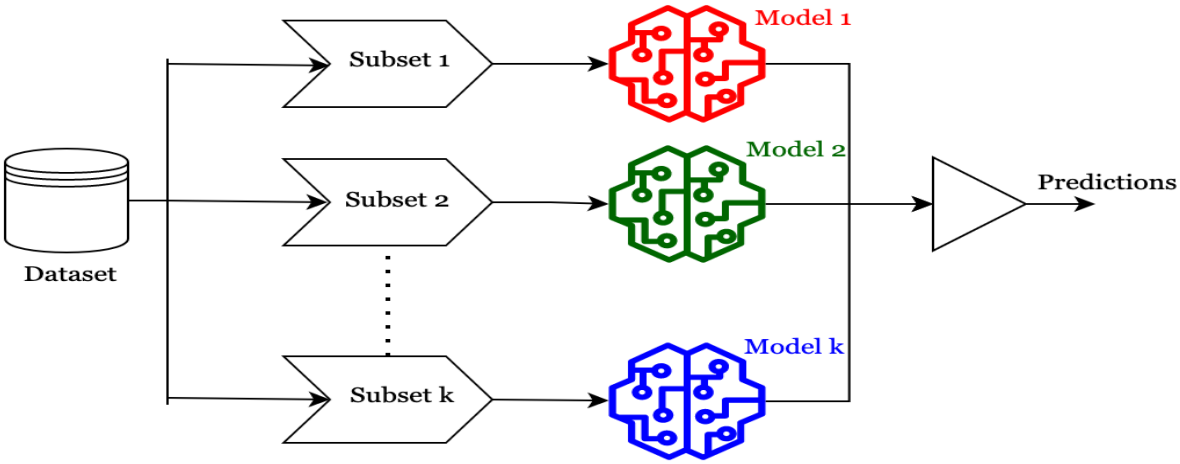
This is a general framework that assigns higher cost to misclassifications of minority class samples. Instead of minimizing overall error, the model minimizes total cost, which matches many real-world problems where missing rare events is more expensive than false alarms.

6.2.2 Adjusting Class Weights

Class weighting is a practical implementation of cost-sensitive learning. It increases penalty for minority errors, making the algorithm pay more attention to the minority class without changing the data distribution.

6.2.3 Ensemble Learning Methods

- **Balanced Random Forest:** Each tree is trained on balanced bootstrap samples; this reduces bias and improves minority detection.
- **EasyEnsemble & BalanceCascade:** Train multiple models on different under-sampled subsets of the majority class and combine predictions, improving robustness while preserving information across subsets.



6.3 Hybrid Approaches

Hybrid approaches combine data-level and algorithm-level methods to benefit from both: better class representation plus better learning focus. These are especially useful in severe imbalance where one approach alone is not enough.

6.3.1 Hybrid Examples

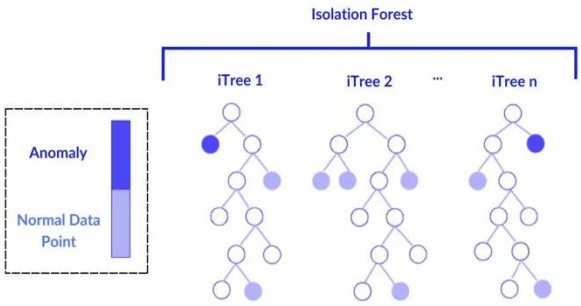
- **SMOTE + XGBoost:** Balances minority data first, then uses boosting to learn complex patterns and improve minority prediction.
- **SMOTE + Random Forest:** Improves minority representation and uses ensemble learning to reduce overfitting.
- **SMOTE + Logistic Regression:** Helps a simpler model learn a fairer boundary after balancing.
- **Under-sampling + Ensemble Models:** Reduces majority data while maintaining robustness through ensemble combination.
- **Class Weighting + Gradient Boosting:** Applies cost penalties during boosting to focus on minority errors.

6.4 Advanced Machine Learning Techniques

Anomaly detection is mainly useful when imbalance is extreme or when minority labeled data is very limited. Instead of learning both classes, the model learns "normal" behavior and flags deviations as anomalies.

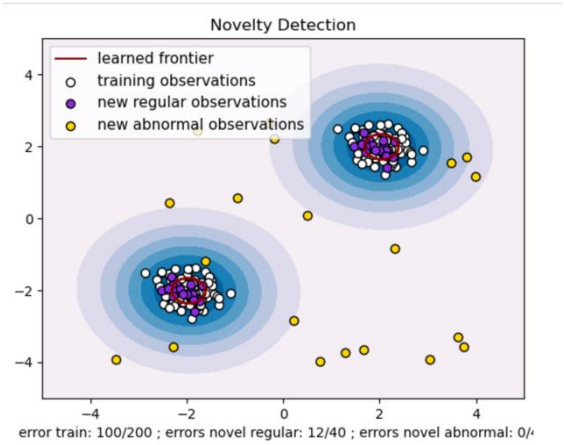
Isolation Forest

Isolates rare observations through random splits; anomalies are easier to isolate.



One-Class SVM

Learns a boundary around normal data and marks points outside as anomalies.

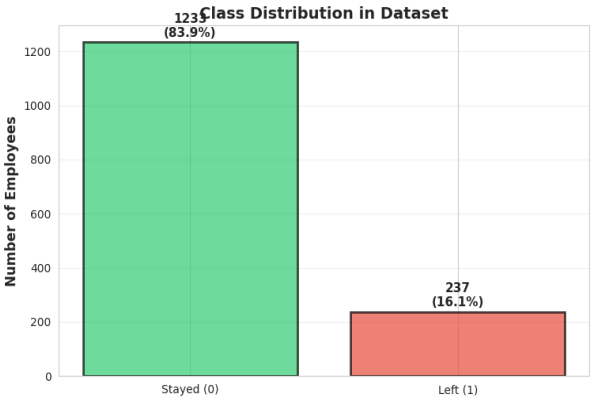


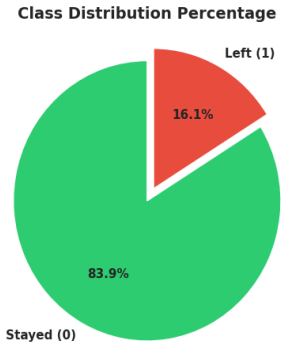
These are typically unsupervised or semi-supervised methods and are most suitable for fraud detection, intrusion detection, and rare-event monitoring where abnormal cases are rare and distinct.

7. Employee Attrition Case Study (Our Mini Project)

To demonstrate the impact of imbalance and improvement through balancing, [1] *"Employee Attrition & Retention Analytics Dataset"* was used.

The task is to predict whether an employee will leave the company based on features such as age, salary, job satisfaction, and work-life balance. The dataset is imbalanced: **Stayed = 1233** and **Left = 237**.





7.1 Experimental Setup

Two models were compared:

- **Model 1 (Imbalanced):** Random Forest trained on the original imbalanced dataset.
- **Model 2 (Balanced):** Apply SMOTE-Tomek to balance classes, then train Random Forest.

7.2 Key Findings

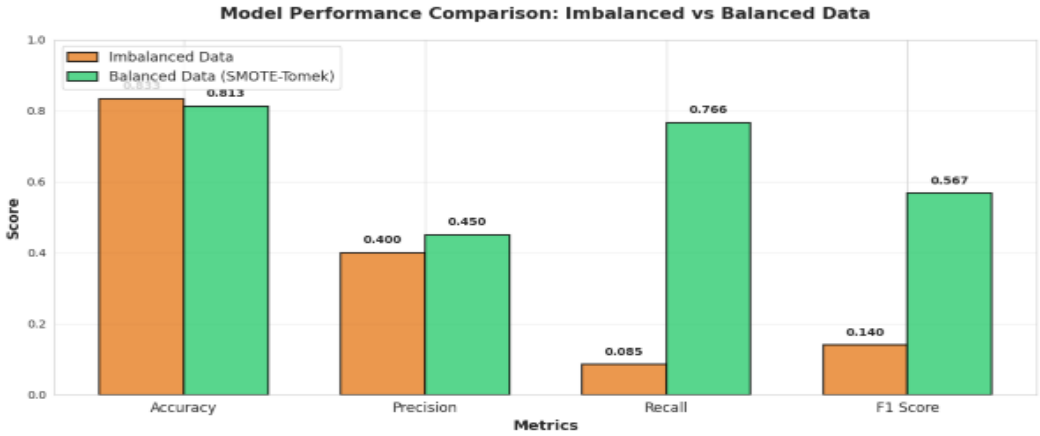
The imbalanced model achieved high accuracy but failed to detect employees who leave. After balancing, minority detection improved substantially.

Results before & after SMOTE-Tomek (Minority class = "Left"):

- Accuracy: **0.833** → **0.813** (slight drop)
- Precision: **0.400** → **0.45** (improvement)
- Recall: **0.085** → **0.766** (major improvement)
- F1-score: **0.14** → **0.567** (major improvement)

This demonstrates a key lesson: accuracy can decrease slightly after balancing, but the model becomes far more useful because it identifies minority cases much better.

[1] Chintawar, A. (n.d.). *Employee Attrition & Retention Analytics Dataset*. Kaggle. https://www.kaggle.com/datasets/ajinkyachintawar/employee-attrition-and-retention-analytics-dataset



7.3 Business Impact

Better detection of at-risk employees enables proactive HR interventions, improves retention strategies, and reduces recruitment and training costs caused by employee turnover.

8. Final Conclusions

8.1 Effective Strategies

Handling class imbalance is essential for producing reliable models. Standard accuracy can be misleading, so models should be evaluated using metrics such as precision, recall, and especially F1-score. Resampling methods, cost-sensitive learning, and ensemble/hybrid approaches significantly improve minority class detection and lead to more meaningful performance in real-world applications.

8.2 Future Directions

Future work can explore hybrid and advanced methods such as ensemble-based hybrids, adaptive resampling, and cost-sensitive loss functions in deep learning. Improving evaluation strategies for imbalanced datasets can further support robust decision-making in complex real-world problems.

